



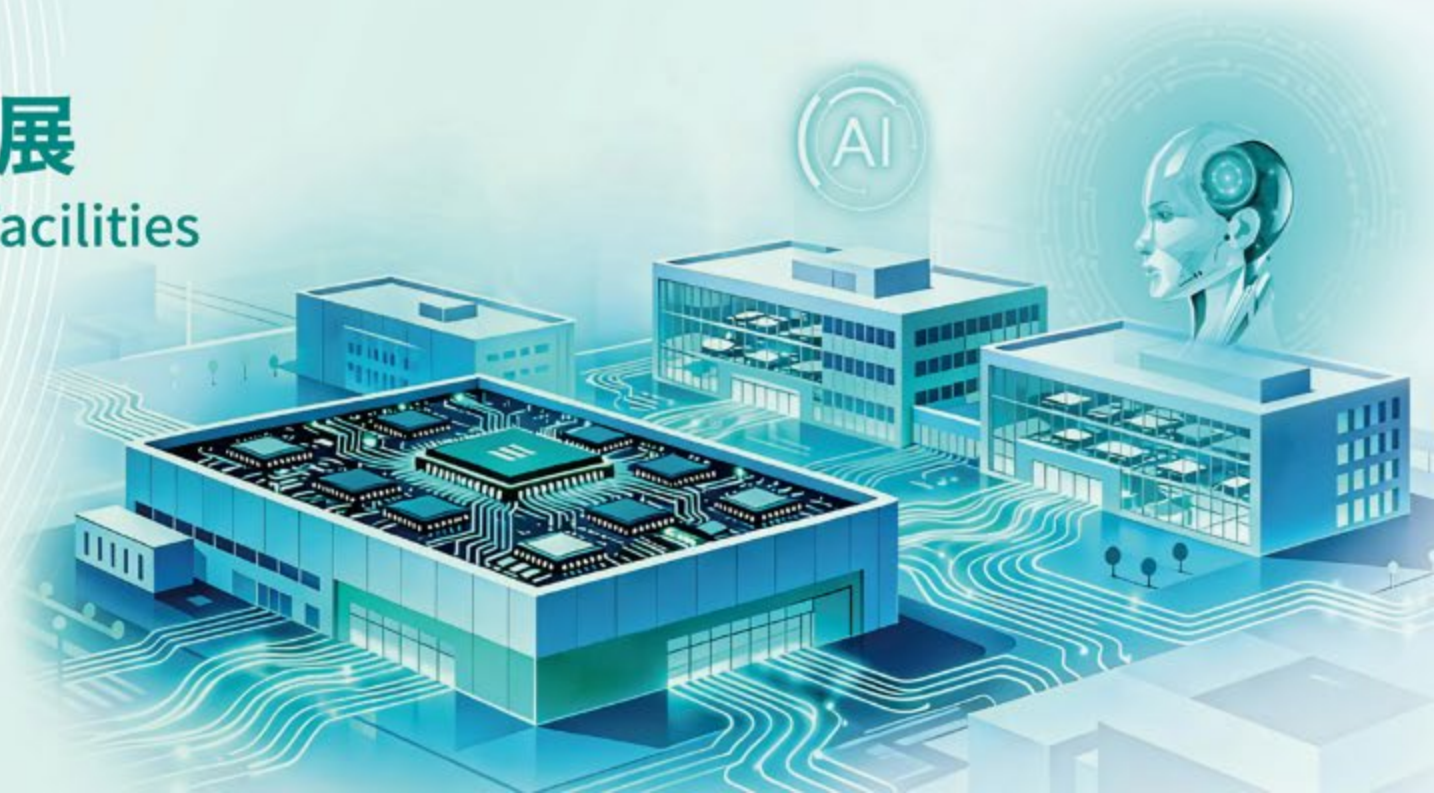
2026 高科技廠房設施國際論壇

High-Tech Facility International Forum 2026

以AI驅動創新加速 高科技廠房設施的未來發展

Accelerating the Future of High-Tech Facilities
Through AI-Driven Innovation

Thursday, September 3, 2026



PHYSICS-CONSTRAINED AI DIGITAL TWIN

FOR ENVIRONMENTAL RISK PREDICTION IN HIGH-TECH FACILITIES

From Passive AMC Monitoring
to Predictive Environmental Intelligence



MONITOR
Detect



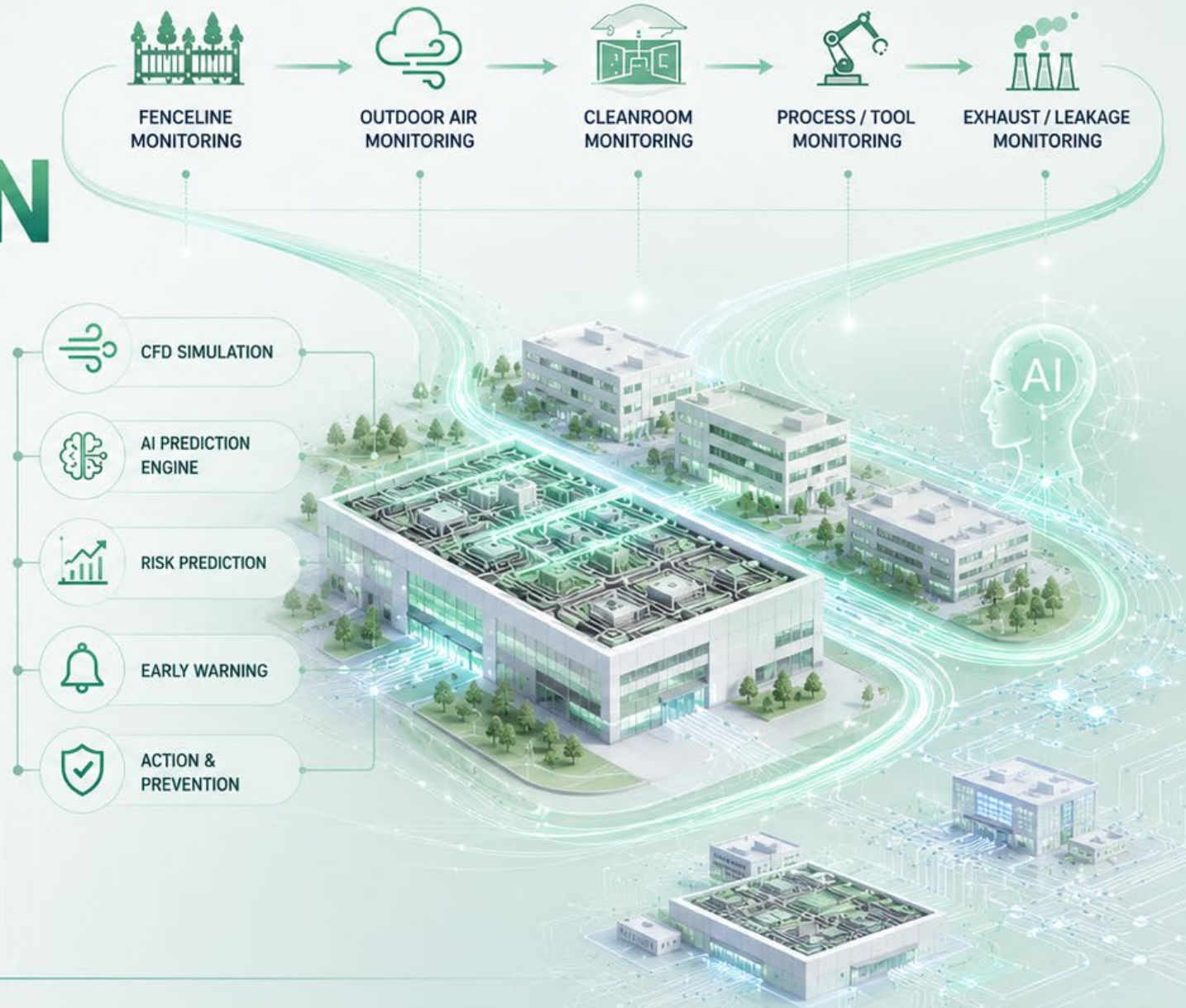
UNDERSTAND
Simulate



PREDICT
Anticipate



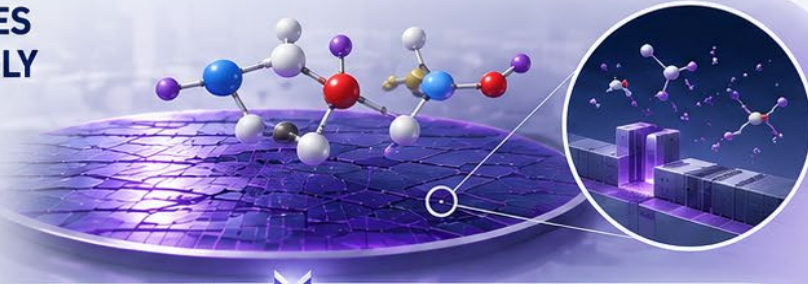
PREVENT
Protect



Semi Fab Environmental Risk – AMC Management

1

ADVANCED NODES
ARE INCREASINGLY
SENSITIVE TO
MOLECULAR
CONTAMINATION



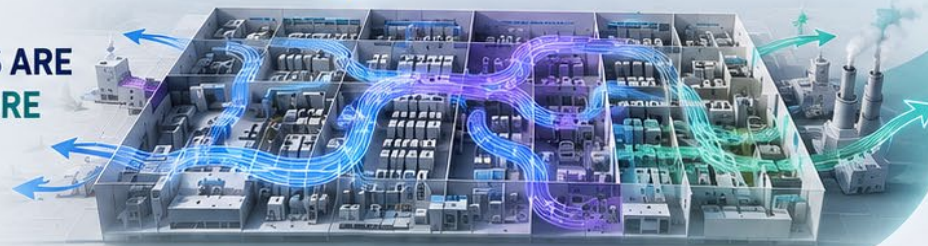
2

FACILITY SCALE
IS GROWING
DRAMATICALLY



3

AIRFLOW
INTERACTIONS ARE
BECOMING MORE
COMPLEX



FROM REACTIVE
MONITORING TO
PREDICTIVE INTELLIGENCE

Integrated Environmental Intelligence
for the Next Generation of Smart Fabs



COMPLETE VISIBILITY

Across the entire
contamination pathway



INTELLIGENT PREDICTION

AI + Physics for accurate
risk forecasting



PROACTIVE PREVENTION

Early warning enables
preventive actions



ESG & COMMUNITY IMPACT

Protect both fab and
surrounding communities

AMC Management – Common Practice

Paint Points



Physics-constrained AI for Fab AMC Management

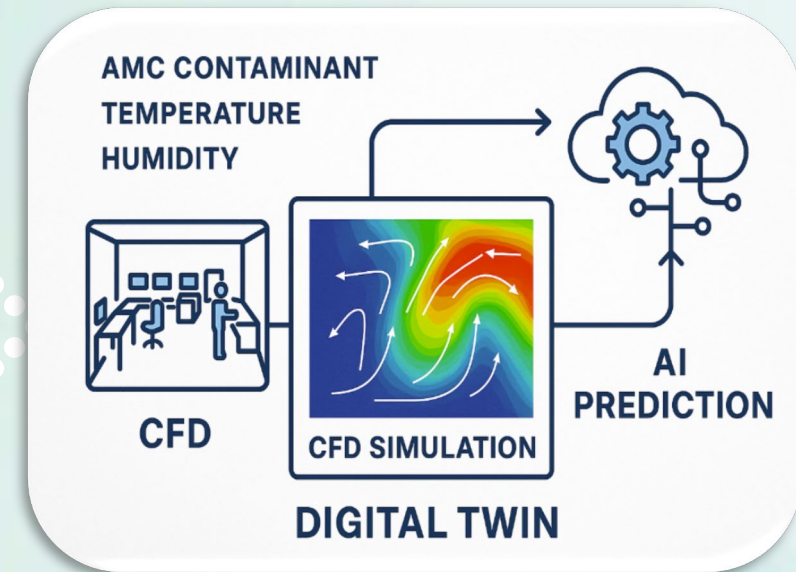
Vision: Environmental Intelligence Platform

From

- Sensors/ Analyzers
- 24/7 Monitoring
- Alarm & React

To

- Sensor Network
- Facility Digital Twin
- CFD Airflow Model
- AI Prediction Engine
- Real-time Risk Dashboard
- Preventive Action System



Why Physics-constrained AI?



RULE-BASED ARCHITECTURE



IF IPA > 20 ppb
AND Wind Speed > 2 m/s



THEN



Alarm



IF NH₃ > 5 ppb



THEN

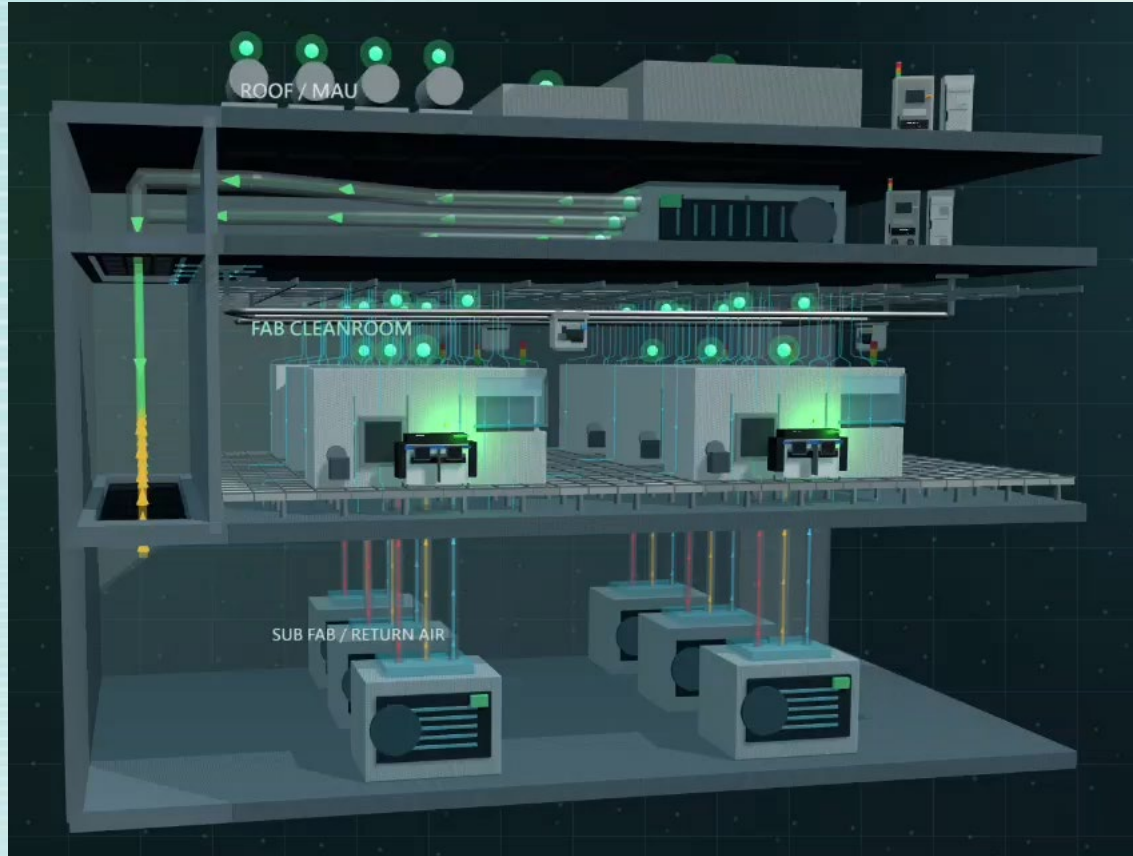


Send Email



What
should happen?

Step 1- Establish Monitoring Network



End-to-End Monitoring Coverage

Fenceline



MAU / Outdoor Air



Cleanroom



Process Tool



Exhaust / Leakage Source

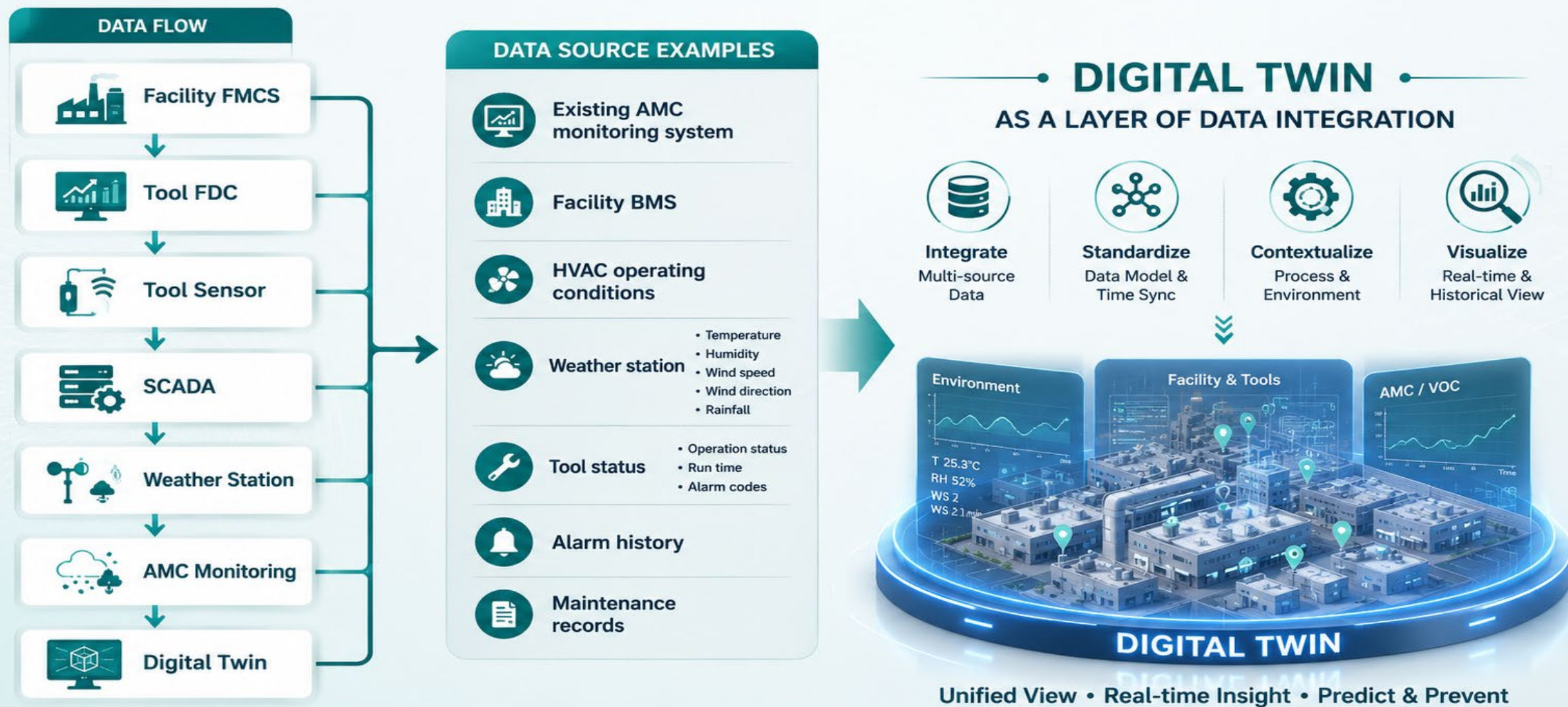


Sampling Points

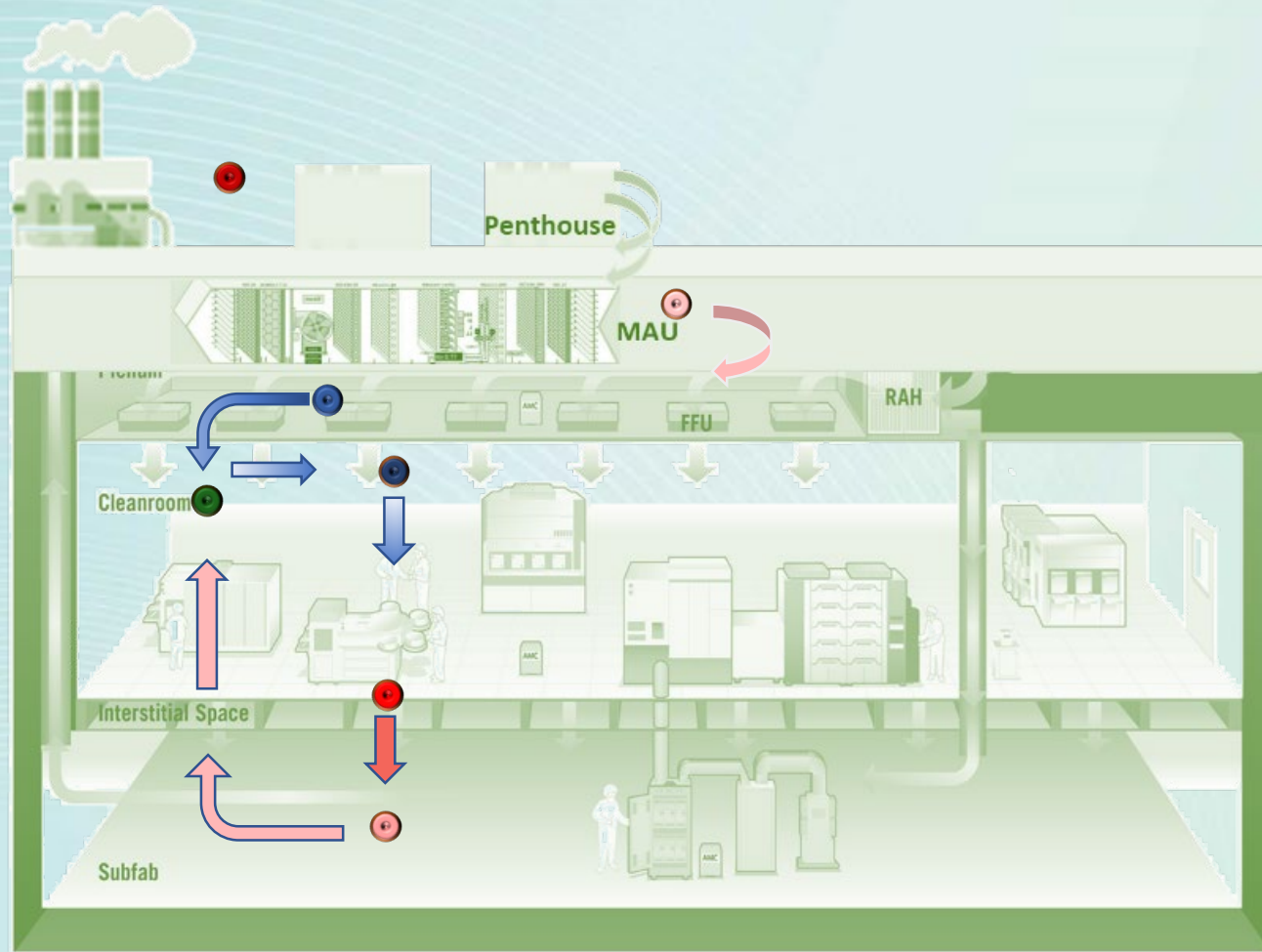
**A unified monitoring strategy
across the entire contamination pathway.**



Step 2- Data Integration



Step 3- Digital Twin Establishment – CFD Simulation (1/2)



● Sampling Points in Plant

Action:
Incorporate Multidimensional Data

Purpose: Develop a DT for
Simulation of Contamination Airflow



Step 3- Digital Twin Establishment – CFD Simulation (2/2)

Action:

1. Identify the Subject
(AMC/Particle/Temp./
Humidity, etc.)
2. Tap in Specific Factors
(Diffusion
coefficient/Seasonal
variables, etc.)

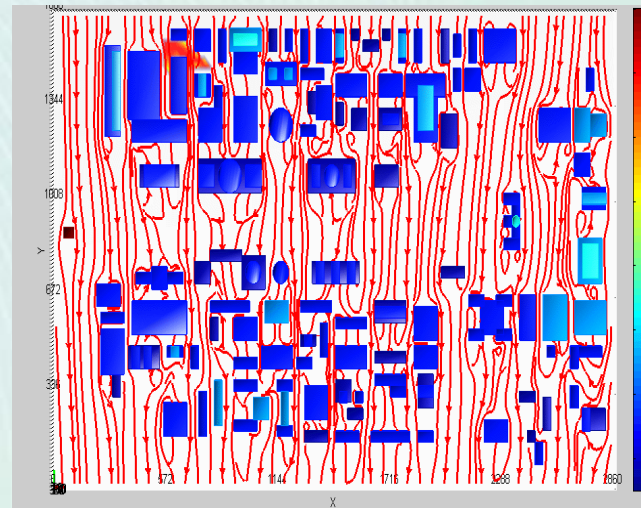
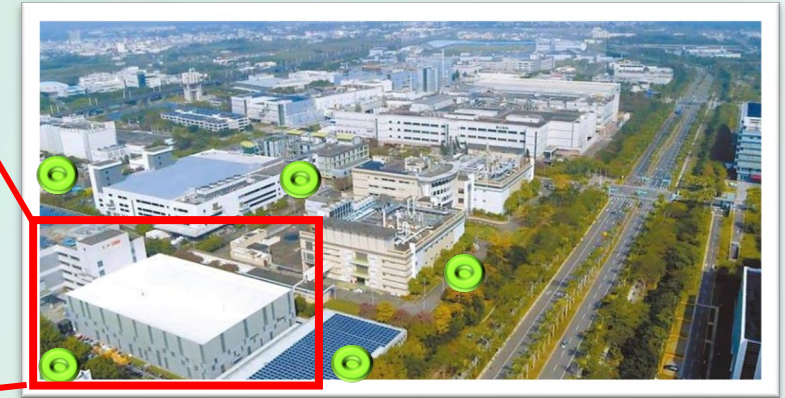
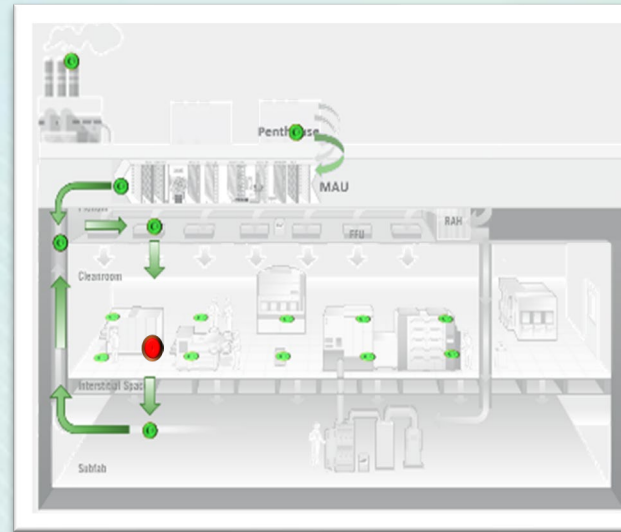
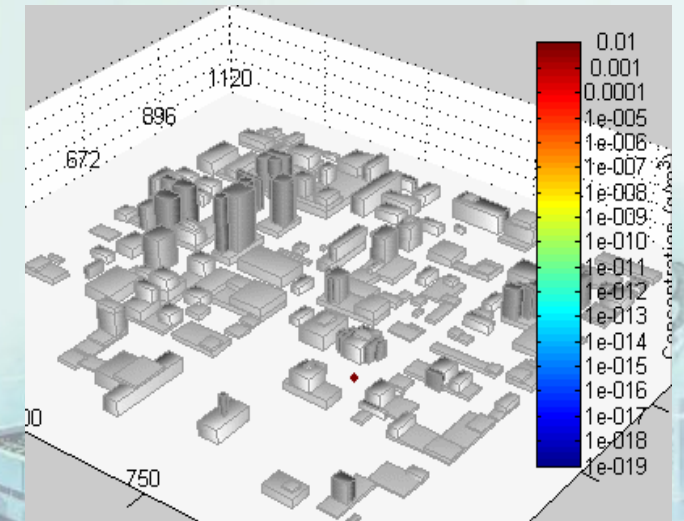
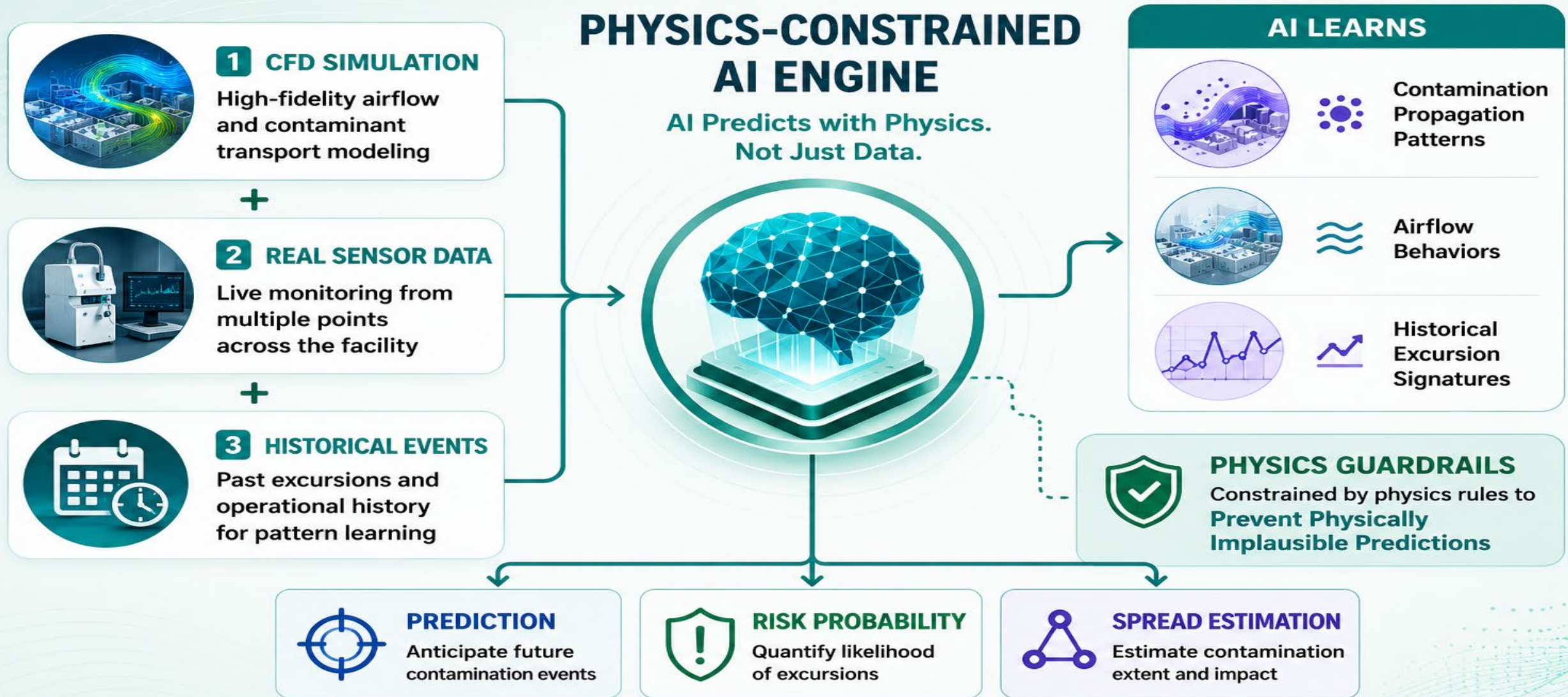


Illustration of Dispersion Modeling

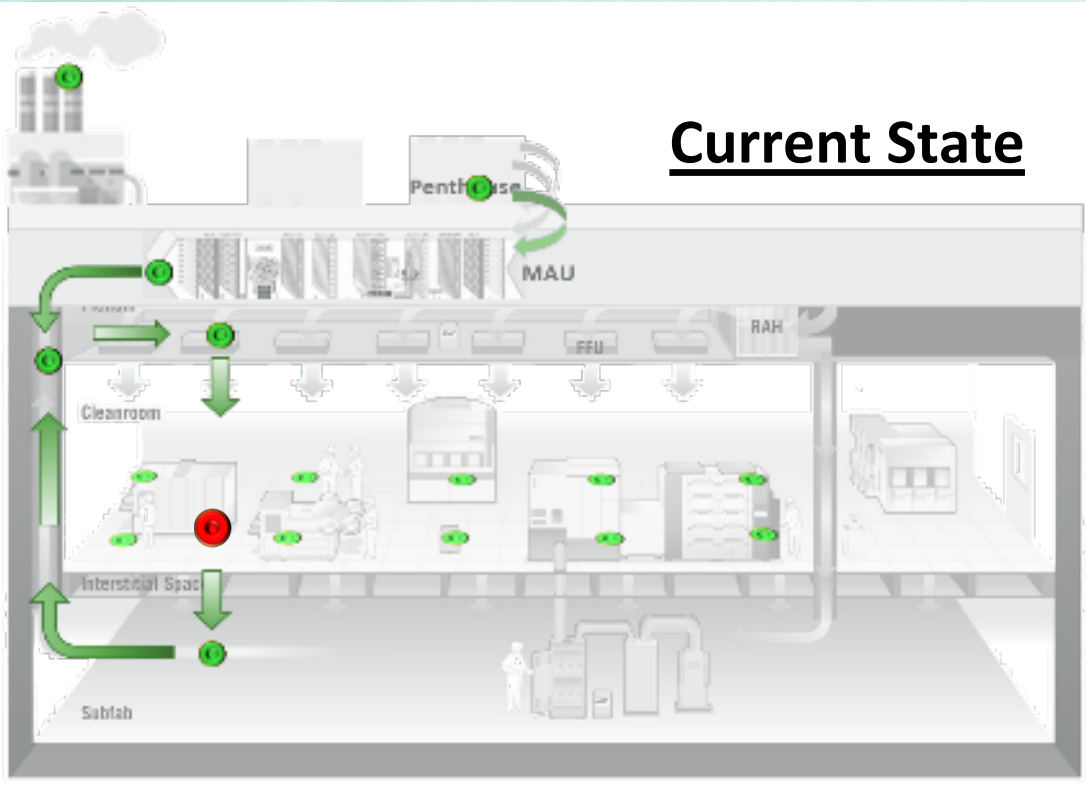


Step 4- Utilize Physics-Constrained AI Solution

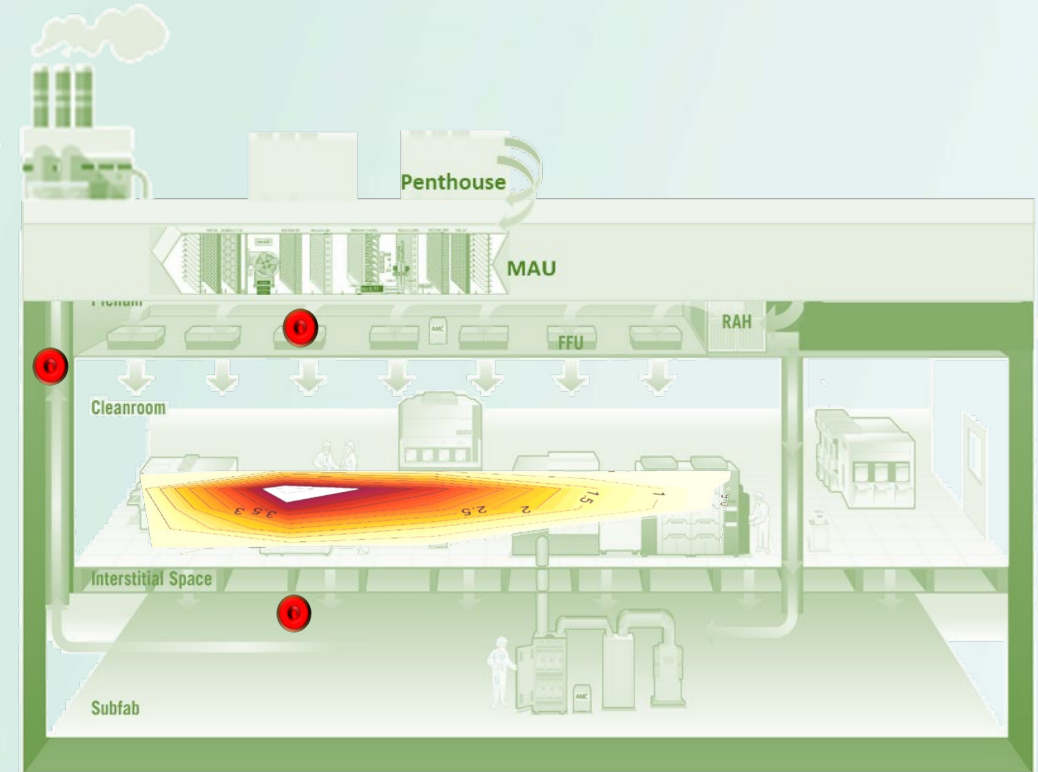


Step 4- Environment Risk Prediction Enabled by AI

Current State



Prediction



30mins later

REAL-TIME PREDICTION DASHBOARD

AI-Powered Environmental Risk Early Warning System



14:35:42
2026/05/20



28°C
65% RH

SYSTEM STATUS

● OPERATIONAL

RISK OVERVIEW

OVERALL RISK LEVEL

HIGH

ACTIVE ALARMS

3

MONITORED POINTS

128

PREDICTION COVERAGE

100%

COUNTDOWN RISK
(EARLIEST RISK ARRIVAL)

CR-3	Cleanroom 3F - Litho	02:15 min
CR-7	Sub Fab 2F - Acid Room	05:42 min
CR-12	MAU 4F - Outdoor Intake	08:37 min
CR-2	Exhaust Duct Area	12:18 min

TOP PREDICTED CONTAMINANTS

Contaminant	Max Predicted Conc.	Trend
IPA	45.3 ppb	↑
TMAH	32.1 ppb	↑
NH ₃	18.7 ppb	→
HF	12.4 ppb	↑
HCl	8.6 ppb	↓

ALARM SUMMARY



	Critical	1
	High	2
	Medium	0
	Low	0

FACILITY MAP - REAL-TIME ENVIRONMENTAL RISK

- RISK HEATMAP LOW HIGH -> AIRFLOW VECTOR

4F

MAU

MAU Room
Outdoor Intake

3F

CLEANROOM

Production Area
Litho / Etch / CMP

2F

SUB FAB

Chemical Room
UPW / HVAC

1F

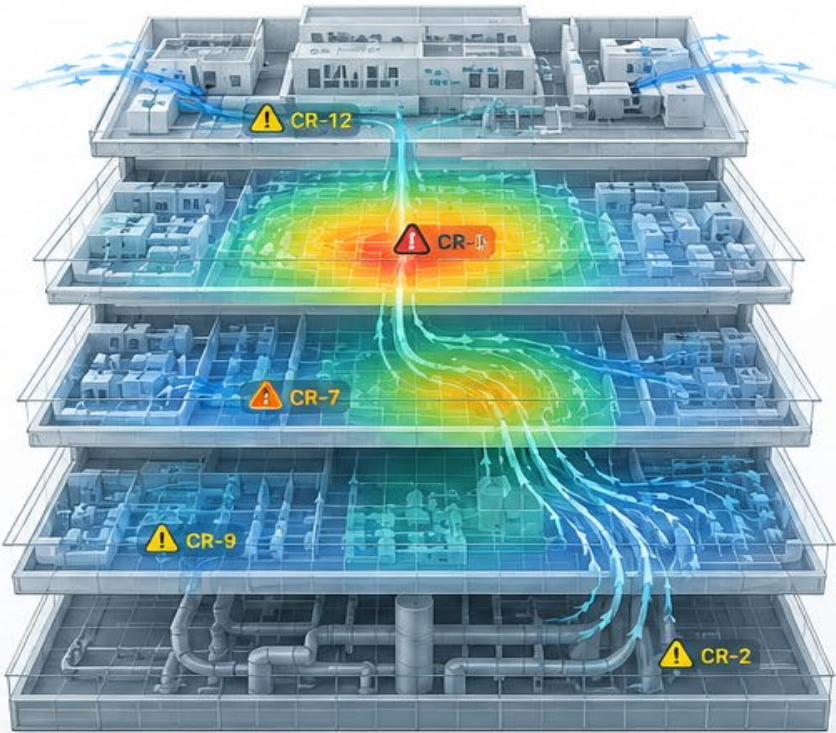
SUB FAB

Utilities / Support

B1

EXHAUST

Exhaust Duct
Scrubber

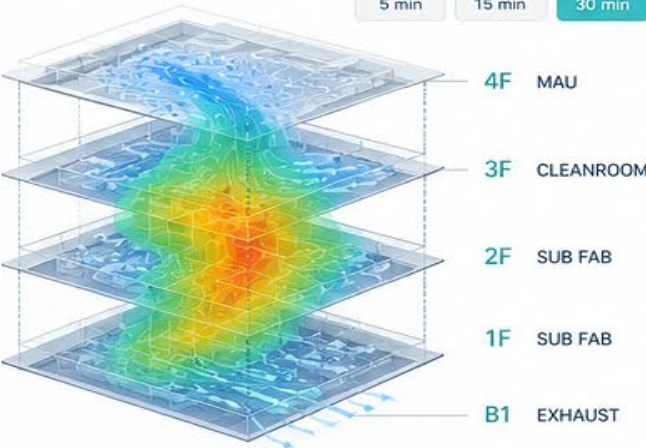


3D



PREDICTED SPREAD (NEXT 30 MIN)

5 min 15 min 30 min



PREDICTION TIMELINE (IPA)



ENVIRONMENTAL CONDITIONS



TEMP
28.0°C



HUMIDITY
65 %



PRESSURE
1012 hPa



WIND
SW 2.4 m/s



RAIN
0 mm

RECENT EVENTS

- 14:33 High risk predicted at Cleanroom 3F - Litho (CR-3)
- 14:31 IPA concentration rising at Cleanroom 3F (CR-3)
- 14:28 MAU 4F intake NH₃ level elevated
- 14:25 System calibration completed
- 14:20 Normal operation

ALARM ZONES

CRITICAL 1 Zone

CR-3
Cleanroom 3F - Litho
♦ IPA | 45.3 ppb



HIGH 2 Zones

CR-7
Sub Fab 2F - Acid Room
♦ TMAH | 32.1 ppb



CR-9
Sub Fab 1F - Utilities Area
♦ VOC | 25.7 ppb



WATCH / CAUTION 1 Zone

CR-2
Exhaust B1 - Duct Area
♦ HF | 12.4 ppb



AI-generated image for illustrative purposes only



Dashboard



Monitoring



Prediction



Analytics



Alerts



Reports



Settings

Real Case Example

Example Scenario: IPA Leakage Scenario
Upon IPA leakage

Conventional Practice/System:

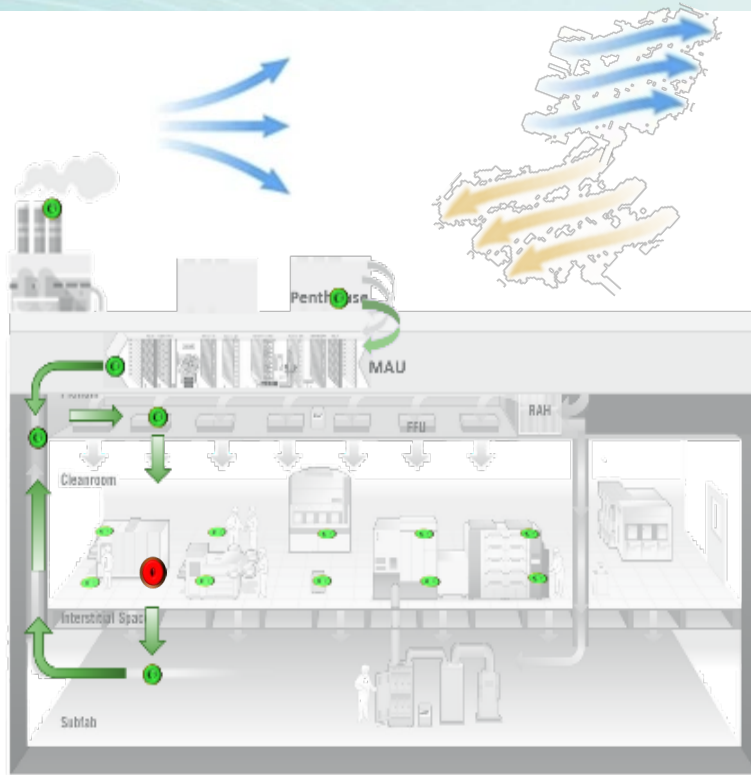
1. alarm **after** concentration rises

TRICORNTech's Physics-Constrained AI Digital Twin:

1. **Predicts** spread direction
2. **Estimates** affected zones
3. **Foresee** concentration trend
4. **triggers** early warning



Scaling – Inside Out → Outside In



- Air travels from **inside of fab to outdoor environment** and **vice versa**.
- **Adjacent facilities/communities** can be the **exposed receptor** as well as the **source** of contaminants to the semi. manufacturing.

Value to Smart Facilities

Monitoring



Prediction



Prevention

Key Benefits

- Yield Protection
- Faster Root Cause Analysis
- Reduced Downtime
- Lower Chemical Risk
- Reduced Human Dependency
- Faster Incident Response



ESG-Driven Env. Intelligence for Semiconductor Fabs

ESG-driven

Shift from compliance monitoring to ESG-driven environmental accountability — fabs are now expected to proactively *prevent* AMC pollution (fab + surrounding communities) as part of environmental justice commitments.

AI-Integrated

Integrate AMC sensing with facility-wide airflow simulation and AI forecasting — enabling fabs to correlate process emissions with airflow behavior, worker exposure zones & external impact — unlocking “predict–prevent–prove” strategy.

Sustainability-Embedded

Position AMC control as part of long-term ESG value & carbon / waste reduction — embedding sustainable design and intelligent control into fab operations to reduce environmental impact while maintaining yield competitiveness.



THANK YOU

BUILDING PREDICTIVE ENVIRONMENTAL
INTELLIGENCE FOR A SAFER TOMORROW



MONITOR
Detect



UNDERSTAND
Simulate



PREDICT
Anticipate



PREVENT
Protect

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